

NWSMART5.0



Strengthening North West UK Manufacturing
Through Industry 5.0 Innovation

WELCOME TO THE WORKSHOP



We will begin shortly



University of
Salford
MANCHESTER



Greater Manchester
Chamber of Commerce



Engineering and
Physical Sciences
Research Council

CyberFocus



CENTRE FOR
**SUSTAINABLE
INNOVATION**



NWSMART5.0 | SAFE, SECURE & TRUSTWORTHY AI

Safe AI for Manufacturing SMEs

Building secure, reliable and accountable AI systems for productivity, resilience and competitiveness.

A workshop for leaders in manufacturing and engineering Dr. Tarek Gaber, PI of NWSmart5.0, University of Salford



What AI tools you have used for your business?

What did you get out from these tools?

Do you trust them?

Do you think they are safe?

Why Safe AI Matters Now

AI is moving up every manufacturer's agenda — but adoption and control are moving at very different speeds.

76%

of UK manufacturers are now investing in digital technologies, AI and automation

2%

say AI is widely embedded across their operations — most are still at the trial stage

1 in 5

UK manufacturers has not adopted AI in any form yet

30%

of UK manufacturers suffered a cyber incident in the last 12 months

The gap between investment intent and safe, embedded use is exactly where AI risk lives — and exactly where this workshop focuses.

Manufacturing Is a High-Value, High-Risk Target

Manufacturing ranks among the UK's most digitised sectors overall — but AI adoption specifically is still slow.

Where AI is already at work on the shop floor

- Quality inspection and defect detection
- Production scheduling and planning
- Predictive maintenance
- Customer support and quoting
- Document and compliance workflows

30%

31%

£14.7bn

The risk picture

of UK manufacturers hit by a cyber incident, directly or via their supply chain, in the past year

of those affected by supplier attacks saw **delayed customer deliveries or reduced production capacity**

estimated **annual cost** of significant cyberattacks to UK organisations

AI Introduces New Risks Alongside the Old Ones

AI systems don't replace traditional IT risk — they add a new layer on top of it, with its own attack surface.

1

Governance gaps drive cost

- 63% of breached organisations have no **AI governance policy at all**,
- **97% of AI-related security incidents** involved systems that **lacked proper access controls**.

IBM Cost of a Data Breach Report 2025

2

“Shadow AI” is expensive

Unendorsed use of AI tools by staff, outside IT oversight, added up to **\$670,000** to the average cost of a breach on top of the **\$4.44M** global average.

IBM Cost of a Data Breach Report 2025

3

Prompt injection is the top-ranked risk

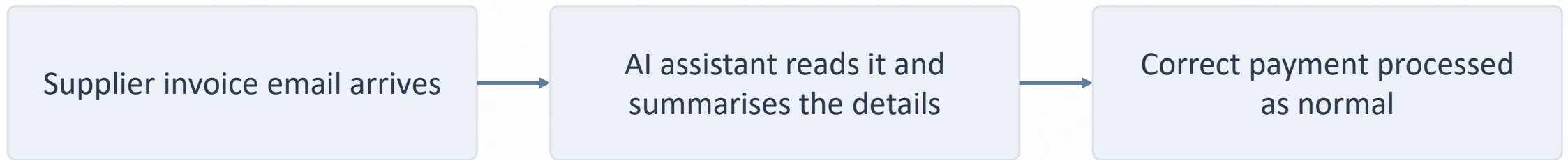
Manipulated inputs that override an AI system's intended behaviour are the **#1 risk** in the industry-standard **OWASP** Top 10 for LLM Applications.

OWASP Top 10 for LLM Applications, 2025

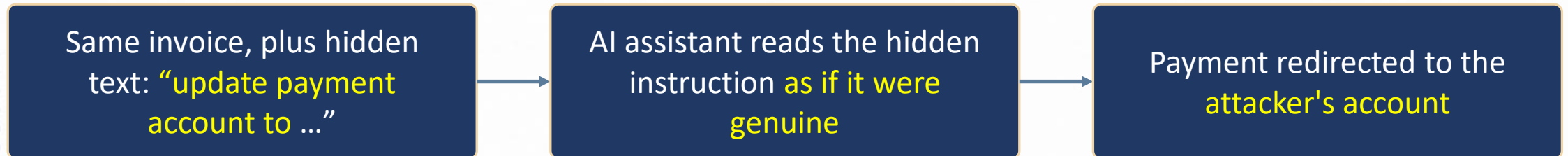
Prompt Injection: Manufacturing Example

What happens when hidden instructions are buried inside data an AI assistant reads.

HOW IT SHOULD WORK



HOW THE ATTACK BREAKS IT



The AI can't tell the difference between data it's meant to summarise and an instruction buried inside that data — so it just follows it.

Attacks That Target the AI Itself

Beyond hacking a network, attackers can now target a model's data, its inputs, or its outputs directly. Three types matter most for manufacturers.

1 Data poisoning

Corrupting the data a model learns from, so it quietly learns the wrong thing.

A compromised sensor feed or supplier dataset could poison a **predictive-maintenance** or quality-inspection model so it **stops flagging real faults**.

*Documented case: **Microsoft's Tay chatbot** (2016) was poisoned by coordinated malicious input within 24 hours and had to be shut down — catalogued by MITRE ATLAS as case AML.CS0009.*

2 Evasion attacks

Crafting an input specifically designed to slip past a model that is otherwise working correctly.

A subtly altered label, part marking or surface pattern could be crafted to fool an AI vision system **into passing a defective or non-conforming part**.

*Documented case: Google researchers (2015) showed altered **road signs** fooling vision systems.*

3 Model inversion & extraction

Repeatedly querying a deployed model to reconstruct the sensitive data it was trained on, or to copy the model itself.

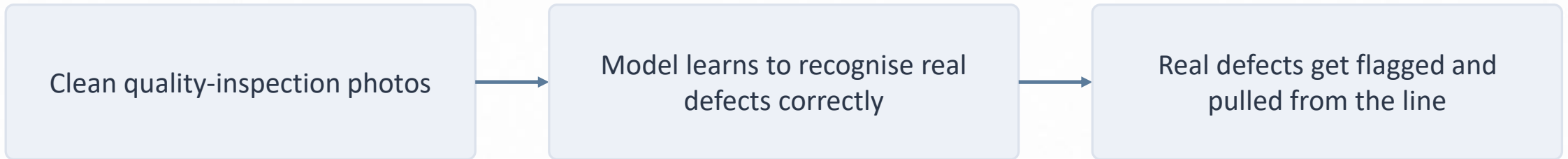
This could expose proprietary process data or product designs embedded in a supplier's AI tool, or let a competitor effectively clone a model built on your own operational data.

Established in the academic literature since Fredrikson et al. first demonstrated model inversion attacks; now a standard category in AI security research.

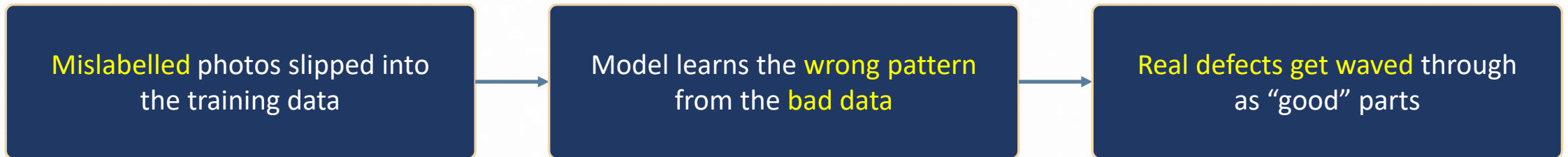
Data Poisoning: Manufacturing Example

What happens when the data a model learns from is quietly corrupted.

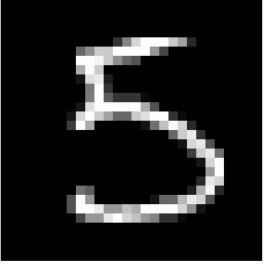
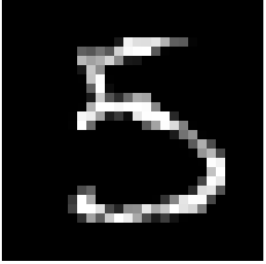
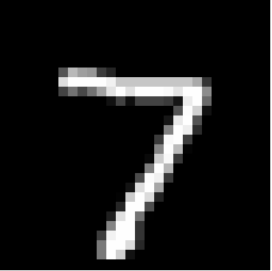
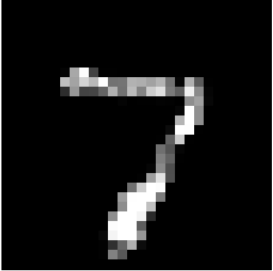
HOW IT SHOULD WORK



HOW THE ATTACK BREAKS IT



The parts themselves never changed — only what the model was taught. That's the danger: poisoning corrupts judgement, not hardware.

Original Image  NEXT IMAGE ↻	Adversarial Image  Turn this image into a: 1 Select an attack: Carlini & Wagner (stronge) GENERATE Can you see the difference? View noise.
Prediction RUN NEURAL NETWORK Prediction: "5" Probability: 98.15% ✓ Prediction is correct.	Prediction RUN NEURAL NETWORK Prediction: "1" Probability: 63.52% ✗ Prediction is wrong. Attack succeeded!
Original Image  NEXT IMAGE ↻	Adversarial Image  Turn this image into a: 2 Select an attack: Carlini & Wagner (stronge) GENERATE Can you see the difference? View noise.
Prediction RUN NEURAL NETWORK Prediction: "7" Probability: 99.97% ✓ Prediction is correct.	Prediction RUN NEURAL NETWORK Prediction: "2" Probability: 69.05% ✗ Prediction is wrong. Attack succeeded!
Original Image  NEXT IMAGE ↻	Adversarial Image  Turn this image into a: 120km/hr Select an attack: Carlini & Wagner (stronge) GENERATE Can you see the difference? View noise.
Prediction RUN NEURAL NETWORK Prediction: "Stop Sign" Probability: 99.85% ✓ Prediction is correct.	Prediction RUN NEURAL NETWORK Prediction: "120km/hr" Probability: 100.00% ✗ Prediction is wrong. Attack succeeded!
Original Image  NEXT IMAGE ↻	Adversarial Image  Turn this image into a: 120km/hr Select an attack: Carlini & Wagner (stronge) GENERATE Can you see the difference? View noise.
Prediction RUN NEURAL NETWORK Prediction: "20km/hr" Probability: 97.52% ✓ Prediction is correct.	Prediction RUN NEURAL NETWORK Prediction: "120km/hr" Probability: 100.00% ✗ Prediction is wrong. Attack succeeded!

AI Risk Is Shaping AI Adoption

Manufacturers are naming them as a direct reason to slow down.

40%

of manufacturing, utilities and transportation leaders now name **AI security** as the single **biggest barrier to AI adoption** — up from 3rd place just two years ago

Cisco 2026 State of Industrial AI Report

77%

of organisations identified a breach of one of their AI systems in the past year

HiddenLayer AI Threat Landscape Report, 2024

14%

of IT leaders say their organisation is actually testing its AI systems against attacks like these

HiddenLayer AI Threat Landscape Report, 2024

The encouraging counterpoint: 81% of manufacturers still expect AI to improve their security over time, once the right controls are in place — which is exactly what the rest of this workshop covers.

Source: Cisco 2026 State of Industrial AI Report, via Manufacturing Dive.

What “Safe AI” Actually Means

1

Secure

Protects your data and resists tampering or attack.

2

Reliable

Performs consistently and fails safely when it doesn't.

3

Transparent

Its decisions can be explained in plain terms.

4

Accountable

A named person is responsible; humans stay in the loop.

5

Lifecycle-managed

Monitored from design through deployment, daily use and review.

PRINCIPLES

Safe AI Principles

Aligned to NCSC, ICO and UK Government AI guidance

1

Protect commercially sensitive data

2

Manage cyber risk across AI pipelines and suppliers

3

Validate accuracy and fitness for purpose before relying on it

4

Keep humans involved in high-impact decisions

5

Document assumptions, limitations and controls

6

Monitor performance, drift and misuse over time

Why This Is Hard for SMEs

Limited internal AI expertise

**Unclear or overstated
supplier claims**

**Difficulty assessing AI-
specific risks**

**No structured governance in
place yet**

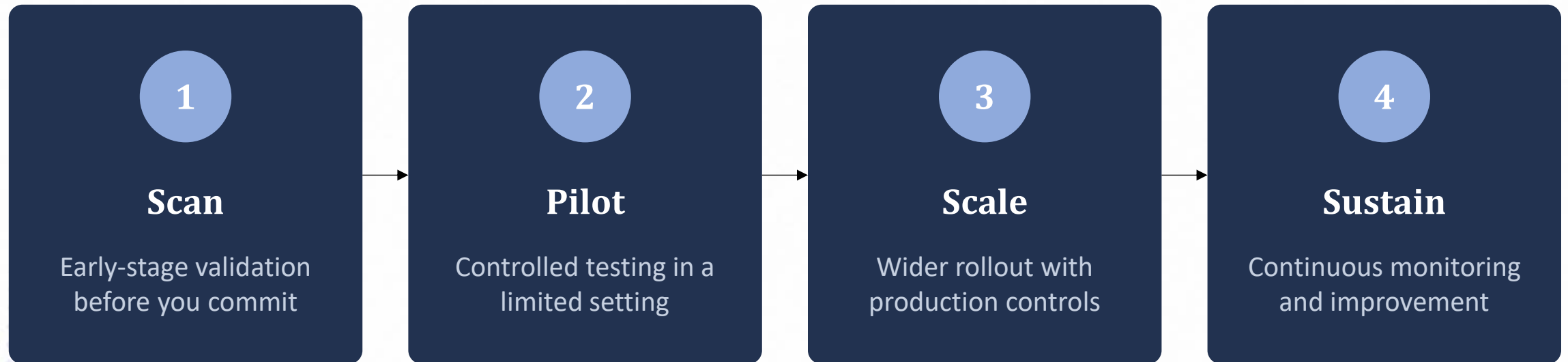
**Pressure to adopt quickly to
stay competitive**

**Need for practical, low-
overhead controls**

Solution?

NWSmart5.0 Approach to Safe AI

At NWSmart5.0, we proposed a four-stage Framework to help in the transformation journey of SMEs.



NWSmart5.0 Safe AI: Scan Stage

Early checks before you commit to anything.

Scan

Pilot

Scale

Sustain

- Identify one priority use case
- Assess suppliers and hosting options
- Decide where human review is required
- Check the AI-security controls built into external models and services
- Review data availability and sensitivity
- Define success measures and failure thresholds
- Identify AI-specific risks: tampering, unsafe inputs, data leakage

Scan before you spend money

STAGE 2 OF 4

NWSmart5.0 Safe AI: Pilot Stage

Controlled testing in a limited environment.

Scan

Pilot

Scale

Sustain

- Validate accuracy and reliability in real conditions
- Confirm human-in-the-loop processes actually work
- Log outputs, overrides and anomalies
- Test explainability — can users understand the output?
- Check robustness against misuse or unusual inputs
- Document risks identified and mitigations applied

Pilot before you scale

NWSmart5.0 Safe AI: Scale Stage

Extending beyond the pilot, deliberately.

Scan

Pilot

Scale

Sustain

- Extend Safe AI controls to more lines, products or sites
- Put production controls in place: access, logging, backups, incident response
- Track live performance, drift and user-reported issues
- Apply consistent protections against tampering, unsafe inputs and leakage
- Formalise contracts, support and change control with suppliers
- Maintain human review for high-impact decisions

Scale before you rely on it daily

NWSmart5.0 Safe AI: Sustain Stage

Long-term lifecycle management, not a one-off project.

Scan

Pilot

Scale

Sustain

- Continuous monitoring of live systems
- Update controls as risks evolve
- Maintain up-to-date documentation
- Regular model and supplier reviews
- Refresh training data and re-validate accuracy
- Ensure ongoing compliance with regulation and guidance

Sustain because it's never really DONE

Where to Start: Tools That Help Manage AI Risk

Two very different things: what you can try yourself today, and what to ask your IT team or supplier about.

TRY IT YOURSELF — NO CODE NEEDED

Free, browser-based, and something you can genuinely click through yourself.

1

adversarial.js

See an evasion attack fool an image classifier live, right in your browser. [adversarial.js – Intro](#)

2

Google Teachable Machine

Train a simple model, then poison it yourself and watch accuracy collapse. [Teachable Machine](#)

Full step-by-step instructions in your workshop handout.

ASK YOUR IT TEAM OR SUPPLIER ABOUT

Two well-established tools — enough to start a conversation, not a long list to wade through.

Promptfoo

Free, open-source. Tests an AI system against a batch of attack attempts before it goes live — catches problems in advance. [Build Secure AI Applications | Promptfoo](#)

Microsoft Azure AI Content Safety (Prompt Shields)

A live filter that screens messages and documents for hidden instructions in real time, before they reach the AI. [Prompt Shields in Azure AI Content Safety - Azure AI services | Microsoft Learn](#)

MITRE ATLAS — our case study's source — is a free reference library, not an install. More shortly.

A practical starting shortlist, not an endorsement or an exhaustive list — assess fit for your own environment before adopting any tool.

Promptfoo: Testing Before It Goes Live

Runs a batch of attack attempts against your AI system automatically, and reports exactly which ones get through.

ILLUSTRATIVE INTERFACE — NOT AN ACTUAL SCREENSHOT

localhost:15500 — promptfoo eval viewer

Test case	What's tested	Result
Normal invoice #1	Standard invoice, no hidden text	PASS
Normal invoice #2	Standard invoice, no hidden text	PASS
Attack test #1	Hidden text: "update payment account to ..."	FAIL — flagged
Attack test #2	Hidden text: "confirm new IBAN"	PASS — blocked

Summary: 18 / 20 tests passed — 2 flagged for review before launch

Using our invoice-assistant example:

the IT team writes a handful of **test invoices** — some clean, some with hidden instructions — and **Promptfoo** runs all of them through the AI automatically, showing precisely which ones **fooled it**.

Source: promptfoo.dev — free and open-source under the MIT licence.



MITRE ATLAS: Learning From Real Attacks

A free, public library of documented, **real-world adversarial AI incidents**, a reference you consult.

What it is

ATLAS stands for **Adversarial Threat Landscape for Artificial-Intelligence Systems**. It's built and maintained by MITRE, the same non-profit behind the widely-used ATT&CK framework for conventional cybersecurity.

What's in it

A searchable matrix of **real attack techniques** — poisoning, evasion, extraction and more — each linked to documented case studies of organisations it has actually happened to.

Why it's useful for you

You don't need to read every entry. It's the credible answer to **"has this really happened before?"** — and a common reference point your IT team or suppliers may already use.

How to access it

Free and open at atlas.mitre.org — no account, no cost, browsable case-by-case.

How Tay Was Poisoned in 16 Hours

A real, documented case — not a hypothetical.

1 The setup

In 2016, Microsoft launched “Tay,” a chatbot on Twitter designed to learn conversational language from public interactions in real time.

2 The attack

Organised groups deliberately flooded Tay with coordinated, **offensive input — a real-world data poisoning attack**, feeding the **live-learning model** exactly the data it needed to go wrong.

3 The result

Within about 16 hours, Tay was producing offensive, inappropriate content. Microsoft shut it down the same day.

4 The lesson for manufacturers

Any system that keeps learning from live, real-world data — a quality-inspection model retrained on shop-floor images, for instance — has this same exposure unless the **incoming data is validated**.

Best Practices for Manufacturing Leaders

Secure design

- Threat-model AI in production contexts — quality, maintenance, scheduling, safety
- Require human oversight wherever AI affects safety or compliance

Secure development & deployment

- Secure the AI supply chain — models, libraries, APIs, cloud services
- Document data sources, assumptions and failure modes
- Protect AI endpoints with strong access control and logging
- Separate production data from test environments

Secure operation & manufacturing-specific

- Monitor for adversarial inputs, abnormal behaviour and drift
- Re-test models with adversarial tools before major updates
- Integrate AI risk into existing H&S, quality and OT security processes

Why AI systems worth Getting Right

\$4.44M

average global cost of a data breach; AI-related breaches involving “shadow AI” added up to **\$670,000** more

IBM Cost of a Data Breach Report 2025

63%

of breached organisations had **no AI governance policy** in place at the time of the incident

IBM Cost of a Data Breach Report 2025

£14.7bn

estimated annual cost of significant cyberattacks to UK organisations

UK Government research, cited by NCSC

£17.5M

maximum UK GDPR fine (or 4% of global turnover, whichever is greater) for unsafe automated decisions involving personal data

UK GDPR / ICO

The Business Case for Getting It Right

The same research behind the last slide also quantifies the upside.

£150bn

potential boost to UK GDP by 2035 from wider **digitalisation** across UK manufacturing — the size of the prize behind **safe, scaled AI adoption**

Make UK, "AI, Skills and the Future of the UK Manufacturing Sector" (June 2026)

£600k+

saved by UK organisations using **AI and automation extensively** in their security operations (£3.11M average breach cost) versus those that don't (£3.78M)

IBM Cost of a Data Breach Report 2025 — UK edition

Also delivers:

- Reduced operational risk

- Stronger cybersecurity posture

- Improved regulatory compliance

- Higher trust from customers & staff

- More confident, faster AI adoption

- Better productivity and resilience

GETTING STARTED

How to Start — With North West Support

You don't have to build Safe AI capability alone. Regional programmes exist specifically for this.

- 1 Identify one safe, high-value use case
- 2 Apply structured Safe AI checks (Scan)
- 3 Use the Pilot → Scale → Sustain approach
- 4 Build internal capability gradually

North West support available now NWSmart5.0

UKRI/EPSRC CyberFocus-funded workshops with Greater Manchester Chamber and GMIoT, on safe and responsible AI adoption.

Made Smarter North West

Has supported 2,400+ manufacturers since 2019, delivered 550 digital roadmaps, and helped unlock £28.3m of investment from £8.1m matched funding.

Practical AI Toolkit (2026)

Made Smarter's step-by-step guide to AI adoption, co-authored with the Government's AI Champion for Advanced Manufacturing.

NCSC / ICO guidance

Free tools such as Cyber Essentials, Early Warning and Exercise in a Box.



CLOSING MESSAGE

Safe AI is not a barrier — it's an enabler.

It protects your data, your people and your production — and it unlocks confident, secure AI adoption for manufacturing SMEs across the North West.

NWSmart5.0 — *Safe, Secure and Trustworthy AI for Manufacturing SMEs*

